Low Nitrogen Fluorescent Microdiamond

Product Sheet

Demanding quantum applications of fluorescent diamond requires NV centers with good spin properties such as long spin coherence (T_2) times and high contrast. Paramagnetic nitrogen centers (P1) can degrade NV spin coherence and diminish performance. Fluorescent diamond particles are usually derived from Type Ib diamond particles containing 100-300 atomic ppm of nitrogen. The Low Nitrogen Fluorescent Microdiamond particles contain sub-50 ppm of nitrogen, and have NV centers with improved spin properties tailored for quantum sensing applications.

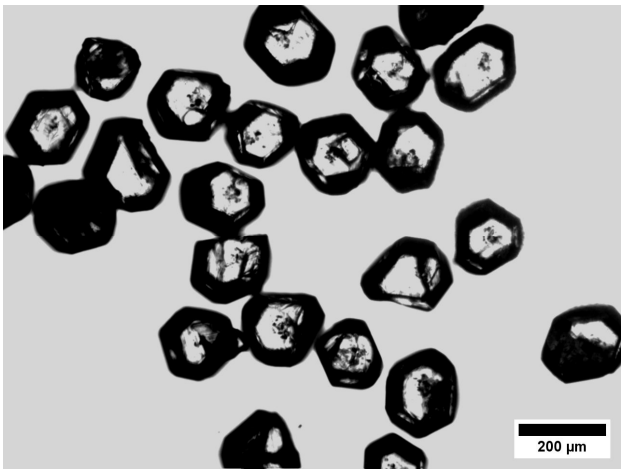


***Figure 1**: Micrograph of HPHT grown microparticles.*

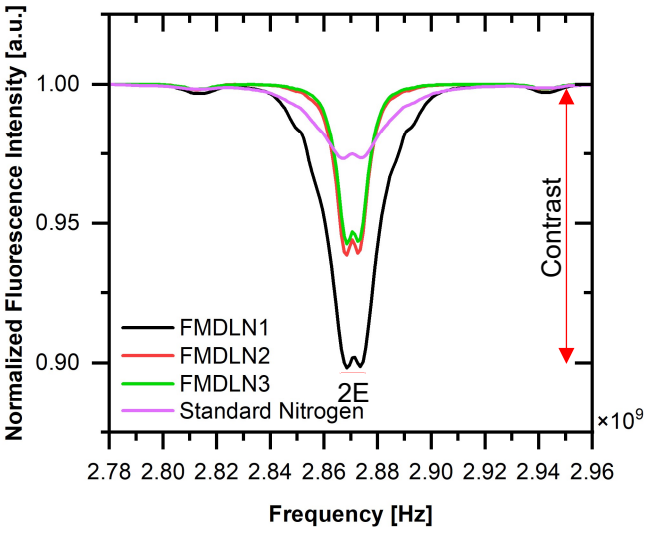


***Figure 2**: Continuous wave ODMR spectrum of FMDNVLN1 sample and a typical HPHT diamond sample with ca. 100 ppm nitrogen content.*

Technical Characteristics

Particle Sizes

Currently available low nitrogen microparticles have nominal sizes of the 150 – 200 µm (**Figure 1**). The particles are synthesized via high pressure high temperature (HPHT) growth.

Nitrogen and NV Content

The nitrogen content was assessed with IR spectroscopy. Particles with three levels of nitrogen content are available. NV content in all samples is estimated to be < 3 ppm.

SKU	Nitrogen Content [ppm]
FMDNVLN1-10mg	20-30
FMDNVLN2-10mg	≤10
FMDNVLN3-10mg	10-20

ODMR and T_1 Relaxation Characterization

Continuous wave ODMR spectra are shown in **Figure 2** and additional summary characteristics are collected in the table below.

SKU	Contrast [%]	E [MHz]	T_1 [ms]
FMDNVLN1-10mg	10.16	4.15	4.89
FMDNVLN2-10mg	5.91	3.05	4.42
FMDNVLN3-10mg	5.57	2.86	4.41
Standard Nitrogen	2.38	5.24	2.17

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ODMR and T_1 Relaxation Characterization

NV centers in reduced nitrogen microparticles exhibit enhanced CW ODMR contrast as well as extended T_1 relaxation times as compared to diamond particles with standard nitrogen content (*ca.* 100 ppm). The FMDLN1 series offers the highest contrast, while the FMDLN2 and FMDLN3 particles exhibit narrower ODMR linewidths (**Figure 2**).

NOTE: ODMR and T1 relaxation characteristics can be subject to measurement configurations and settings. All data presented were collected under green LED excitation providing approximately 3.4 mW/mm^2 incident power density at 10x magnification. Microwave power was kept at -5 dBm (0.32 mW) for collection of ODMR spectra.

Application Areas

- NV Magnetometry
- NV Thermometry

Have additional questions? Contact us at info@adamasnano.com

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